1 Subspaces of a Vector Space

The set $\mathscr{S}(V)$ of all subspaces of a vector space V is partially-ordered by set inclusion. This means that the relationship $U \subseteq W$ is

- reflexive: $U \subseteq U$ for every $U \in \mathscr{S}(V)$.
- antisymmetric: If $U \subseteq W$ and $W \subseteq U$, then $U = W$.
- transitive: If $T \subseteq U$ and $U \subseteq W$, then $T \subseteq W$.

We will say that "U is smaller than W" or "W is larger than U" if U is a proper subset of W. If neither U nor W are subsets of each other, then they are not comparable. Note that $\mathscr{S}(V)$ has a *minimum element* and *maximum element*, namely $\{0\}$ and V, respectively. This just means that $\{0\}$ is contained in every subspace, and that every subspace is contained in V. (Note that the maximum and minimum elements of a partially-ordered set, if they exist, are necessarily unique.)

Given two subspaces U and W of V, we have seen that $U \cap W$ is again a subspace of V.

Proposition 1.1. *Given subspaces U and W of V, $U \cap W$ is the largest subspace of V contained in both U and W.*

Exercise 1.2. Prove Proposition 1.1 by showing that if T is any subspace contained in both U and W, then $T \subseteq U \cap W$.

A virtually identical proof shows that, more generally, if $\{W_\alpha : \alpha \in \mathscr{A}\}$ is any collection of subspaces of V then $\bigcap_{\alpha \in \mathscr{A}} W_\alpha$ is the largest subspace of V contained in each W_α.

We have just answered the question,

"Given two subspaces U and W of a vector space V, what is the largest subspace of V contained in both U and W?" [1]

and found that the answer is the intersection, $U \cap W$.

There is another natural question, dual to the previous one:

"Given two subspaces U and W of a vector space V, what is the smallest subspace of V containing both U and W?"

Exercise 1.3. Explain why this question also makes sense.

[1] Note that this question makes sense, since we know that there is at least one subspace of V contained in both U and W, namely $\{0\}$.

At the level of sets, it is true that any subset of V containing both U and W must also contain $U \cup W$, but, as we have already seen, $U \cup W$ is in general not a subspace of V. We will now show that the answer to this question is, instead, the *sum* of U and W, which we define below.

Definition 1.4 (Sum of subspaces)**.** If U and W are subspaces of a vector space V, then the *sum* of U and W, denoted by $U+W$, is defined to be the set

$$U+W := \{u+w : u \in U \text{ and } w \in W\}.$$

That is, v is an element of $U+W$ if and only if there exists $u \in U$ and $w \in W$ such that $v = u+v$.

Example 1.5. Let $U = \{(a,0) \in F^2\}$ and $W = \{(0,b) \in F^2\}$. Given $(x,y) \in F^2$, we can write $(x,y) = (x,0)+(0,y)$, so we see that $F^2 = U+W$.

Exercise 1.6. Let $U = \{(a,0) \in F^2\}$ and $W = \{(b,b) \in F^2\}$. Show that $F^2 = U+W$.

Proposition 1.7. *If U and W are subspaces of a vector space V, then so is $U+W$.*

Exercise 1.8. Prove Proposition 1.7.

More generally, if $V_1, \ldots, V_n$ is any *finite* collection of subspaces of V, then we define

$$V_1 + \cdots + V_n := \{v_1 + \cdots + v_n : v_1 \in V_1, \cdots, v_n \in V_n\},$$

which is again a subspace of V.

Exercise 1.9. Prove that $U+W$ contains both U and W.

Proposition 1.10. *If U and W are subspaces of V, then $U+W$ is the smallest subspace of V containing both U and W.*

Exercise 1.11. Prove Proposition 1.10.

Similarly, for any finite collection $V_1,\dots,V_n$ of subspaces of a vector space V, $V_1+\cdots+V_n$ is the smallest subspace of V containing $V_1,\dots,V_n$.

Example 1.12. Consider the following subspaces of F^4:

$$\begin{aligned} U &= \{(x,x,y,y) \in F^4\},\\ V &= \{(x,x,x,y) \in F^4\},\\ W &= \{(x,x,y,z) \in F^4\}. \end{aligned}$$

We will show that $W = U+V$. To do so, we need to show that $U+V \subseteq W$ and $W \subseteq U+V$. Adding and element $(a,a,a,b) \in U$ and an element $(c,c,d,e) \in V$ gives

$$(a,a,a,b)+(c,c,d,e) = (a+c,a+c,a+d,a+e),$$

which is of the form (x,x,y,z) with $x,y,z \in F$. This shows that $U+V \subseteq W$.

To show the opposite inclusion, let $(x,x,y,z) \in W$. We need to find $(a,a,b,b) \in U$ and $(c,c,c,d) \in V$ such that

$$\begin{aligned} (x,x,y,z) &= (a,a,b,b)+(c,c,c,d)\\ &= (a+c,a+c,b+c,b+d). \end{aligned}$$

This leads to the system of equations

$$\begin{aligned} a+0+c+0 &= x,\\ 0+b+c+0 &= y,\\ 0+b+0+d &= z. \end{aligned}$$

Row-reducing the corresponding augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & x\\ 0 & 1 & 1 & 0 & y\\ 0 & 1 & 0 & 1 & z \end{array}\right] \mapsto \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & x-y+z\\ 0 & 1 & 0 & 1 & z\\ 0 & 0 & 1 & -1 & y-z \end{array}\right],$$

we find an infinite number of solutions

$$\begin{aligned} a &= -t+x-y+z\\ b &= -t+z\\ c &= t+(y-z)\\ d &= t. \end{aligned}$$

For instance, when $t=0$ we have

$$(x,x,y,z) = (x-y+z,x-y+z,z,z)+(y-z,y-z,y-z,0).$$

Thus, $W \subseteq U+V$ and therefore $W = U+V$.

Exercise 1.13. Consider the following subspaces of F^3:

$$V_1 = \{(x,y,0) \in F^3\},$$
$$V_2 = \{(0,0,z) \in F^3\},$$
$$V_3 = \{(0,y,y) \in F^3\}.$$

Show that $F^3 = V_1 + V_2 + V_3$.